In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [4]:
```
df = pd.read_csv("C:/Users/Ganesh/Downloads/HousingData.csv")
```

In [5]:
```
df.head()
```

Out[5]:

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	I
0	0.00632	18.0	2.31	0.0	0.538	6.575	65.2	4.0900	1	296	15.3	396.90	
1	0.02731	0.0	7.07	0.0	0.469	6.421	78.9	4.9671	2	242	17.8	396.90	
2	0.02729	0.0	7.07	0.0	0.469	7.185	61.1	4.9671	2	242	17.8	392.83	
3	0.03237	0.0	2.18	0.0	0.458	6.998	45.8	6.0622	3	222	18.7	394.63	
4	0.06905	0.0	2.18	0.0	0.458	7.147	54.2	6.0622	3	222	18.7	396.90	

In [6]:
```
df.tail()
```

Out[6]:

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B
501	0.06263	0.0	11.93	0.0	0.573	6.593	69.1	2.4786	1	273	21.0	391.99
502	0.04527	0.0	11.93	0.0	0.573	6.120	76.7	2.2875	1	273	21.0	396.90
503	0.06076	0.0	11.93	0.0	0.573	6.976	91.0	2.1675	1	273	21.0	396.90
504	0.10959	0.0	11.93	0.0	0.573	6.794	89.3	2.3889	1	273	21.0	393.45
505	0.04741	0.0	11.93	0.0	0.573	6.030	NaN	2.5050	1	273	21.0	396.90

In [7]:
```
df.describe()
```

Out[7]:

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE
count	486.000000	486.000000	486.000000	486.000000	506.000000	506.000000	486.000000
mean	3.611874	11.211934	11.083992	0.069959	0.554695	6.284634	68.518519
std	8.720192	23.388876	6.835896	0.255340	0.115878	0.702617	27.999513
min	0.006320	0.000000	0.460000	0.000000	0.385000	3.561000	2.900000
25%	0.081900	0.000000	5.190000	0.000000	0.449000	5.885500	45.175000
50%	0.253715	0.000000	9.690000	0.000000	0.538000	6.208500	76.800000
75%	3.560263	12.500000	18.100000	0.000000	0.624000	6.623500	93.975000
max	88.976200	100.000000	27.740000	1.000000	0.871000	8.780000	100.000000

In [8]:
```
df.shape
```

Out[8]:
```
(506, 14)
```

In [9]:
```
df.dtypes
```

Out[9]:
```
CRIM       float64
ZN         float64
INDUS      float64
CHAS       float64
NOX        float64
RM         float64
AGE        float64
DIS        float64
RAD          int64
TAX          int64
PTRATIO    float64
B          float64
LSTAT      float64
MEDV       float64
dtype: object
```

In [10]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 506 entries, 0 to 505
Data columns (total 14 columns):
 #   Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0   CRIM     486 non-null    float64
 1   ZN       486 non-null    float64
 2   INDUS    486 non-null    float64
 3   CHAS     486 non-null    float64
 4   NOX      506 non-null    float64
 5   RM       506 non-null    float64
 6   AGE      486 non-null    float64
 7   DIS      506 non-null    float64
 8   RAD      506 non-null    int64
 9   TAX      506 non-null    int64
 10  PTRATIO  506 non-null    float64
 11  B        506 non-null    float64
 12  LSTAT    486 non-null    float64
 13  MEDV     506 non-null    float64
dtypes: float64(12), int64(2)
memory usage: 55.5 KB
```

In [11]:
```
df.isna().sum()
```

Out[11]:
```
CRIM       20
ZN         20
INDUS      20
CHAS       20
NOX         0
RM          0
AGE        20
DIS         0
RAD         0
TAX         0
PTRATIO     0
B           0
LSTAT      20
MEDV        0
dtype: int64
```

In [12]:
```
mean_value = df['CRIM'].mean()
```

In [13]:
```
# Calculate the mean of each numeric column
means = df.mean()
# Impute missing values with the mean values
df.fillna(value=means, inplace=True)
# Check for any remaining missing values
print(df.isnull().sum())
```

```
CRIM       0
ZN         0
INDUS      0
CHAS       0
NOX        0
RM         0
AGE        0
DIS        0
RAD        0
TAX        0
PTRATIO    0
B          0
LSTAT      0
MEDV       0
dtype: int64
```

In [14]:
```
target_feature = 'MEDV'
```

In [15]:
```
x = df.drop(target_feature, axis=1)
y = df[target_feature]
```

In [16]:
```
x.head()
```

Out[16]:

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B
0	0.00632	18.0	2.31	0.0	0.538	6.575	65.2	4.0900	1	296	15.3	396.90
1	0.02731	0.0	7.07	0.0	0.469	6.421	78.9	4.9671	2	242	17.8	396.90
2	0.02729	0.0	7.07	0.0	0.469	7.185	61.1	4.9671	2	242	17.8	392.83
3	0.03237	0.0	2.18	0.0	0.458	6.998	45.8	6.0622	3	222	18.7	394.63
4	0.06905	0.0	2.18	0.0	0.458	7.147	54.2	6.0622	3	222	18.7	396.90

In [17]:
```
y.head()
```

Out[17]:
```
0    24.0
1    21.6
2    34.7
3    33.4
4    36.2
Name: MEDV, dtype: float64
```

In [18]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.2,random_state=0)
```

In [19]:
```
from sklearn.linear_model import LinearRegression
# Create an instance of a LinearRegression() model named regression.
regression = LinearRegression()
```

In [20]:
```
regression.fit(x_train, y_train)
```

Out[20]:
```
▾ LinearRegression
▶ Parameters
```

In [21]:
```
train_score=round(regression.score(x_train,y_train)*100,2)
print('Train score of Linear Regression:',train_score)
```

```
Train score of Linear Regression: 76.49
```

In [22]:
```
print('Coefficients" ', regression.coef_)
```

```
Coefficients"  [-1.26194005e-01  3.76363553e-02 -6.26295345e-02  2.70382928e+00
 -1.45015824e+01  4.08006958e+00 -2.11509464e-02 -1.41798662e+00
  1.96343241e-01 -8.70651696e-03 -1.01396225e+00  8.29504244e-03
 -4.19861039e-01]
```

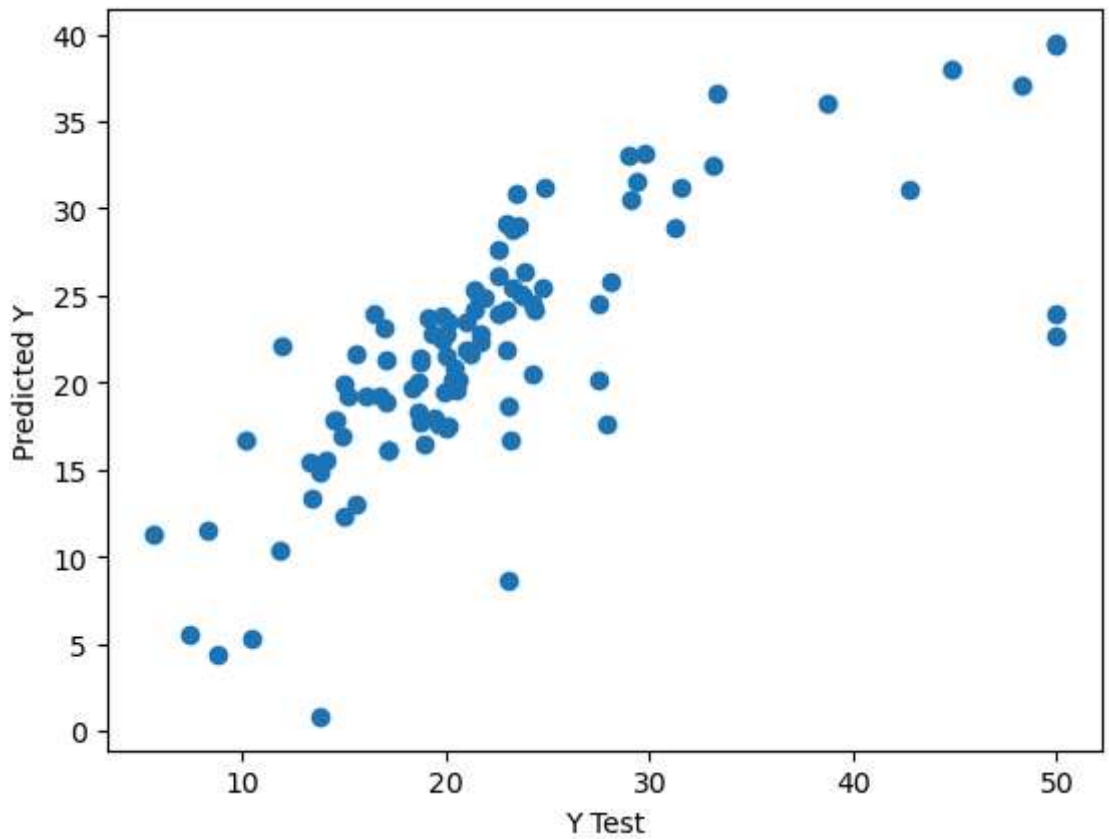
In [23]:
```
predictions = regression.predict(x_test)
```

In [24]:
```
predictions
```

Out[24]:
```
array([26.175296  , 22.64747588, 29.1456294 , 11.52971235, 21.65312134,
       19.42320699, 20.18413017, 21.46914355, 19.1985363 , 19.98228162,
        4.32483046, 16.16891668, 16.87682404,  5.31232373, 39.36827861,
       33.09358732, 21.9152876 , 36.61918436, 31.52676377, 23.52713482,
       24.96022461, 23.69866912, 20.88033802, 30.55074901, 22.74081741,
        8.66805959, 17.65119072, 17.93088633, 36.01223185, 21.16299556,
       17.83464361, 17.43306603, 19.5240167 , 23.50605522, 28.97262793,
       19.21808862, 11.23997435, 23.94256597, 17.86786717, 15.40849806,
       26.3630836 , 21.5193299 , 23.78733694, 14.84041522, 23.9445175 ,
       24.97067627, 20.11366175, 23.08636158, 10.42208266, 24.52832122,
       21.60847326, 18.66228165, 24.53362832, 31.03502944, 12.97457826,
       22.38536236, 21.34822822, 16.10928673, 12.37477824, 22.78596712,
       18.28714824, 21.91802045, 32.49771603, 31.21256855, 17.47867791,
       33.18861907, 19.17896285, 19.94662594, 20.17142015, 23.90228857,
       22.81288844, 24.17911208, 30.83402844, 28.87481037, 25.14581721,
        5.55072029, 37.0183454 , 24.15428003, 27.67587636, 19.63884644,
       28.74874123, 18.83204358, 17.63305678, 37.97947167, 39.49507972,
       24.17228966, 25.33605088, 16.75044819, 25.43224687, 16.65089426,
       16.49186628, 13.37283452, 24.81689254, 31.21188699, 22.0891919 ,
       20.49360168,  0.8229737 , 25.5004737 , 15.5481509 , 17.72901193,
       25.77663998, 22.43131323])
```

In [25]:
```
plt.scatter(y_test, predictions)
plt.xlabel('Y Test')
plt.ylabel('Predicted Y')
```

Out[25]:
```
Text(0, 0.5, 'Predicted Y')
```



In [26]:
```
from sklearn.metrics import r2_score
score = round(r2_score(y_test,predictions)*100,2)
print("r_2 score:", score)
```

```
r_2 score: 57.03
```

In [27]:
```
round(regression.score(x_test, y_test)*100,2)
```

Out[27]: 57.03

In [31]:
```
from sklearn import metrics
print('Mean Absolute Error on test data of Linear Regression: ',metrics.mean_absolu
print('Mean Squared Error on test data of Linear Regression: ',metrics.mean_squared
print('Root Mean Squared Error on test data of Linear Regression: ',np.sqrt(metrics
```

```
Mean Absolute Error on test data of Linear Regression:  3.9616211239591146
Mean Squared Error on test data of Linear Regression:  34.98738954423872
Root Mean Squared Error on test data of Linear Regression:  5.915013909048627
```

In [32]:
```
df1 = pd.DataFrame({'Actual':y_test, 'Predicted':predictions, 'Variance':y_test-pre
df1.head()
```

Out[32]:

	Actual	Predicted	Variance
329	22.6	26.175296	-3.575296
371	50.0	22.647476	27.352524
219	23.0	29.145629	-6.145629
403	8.3	11.529712	-3.229712
78	21.2	21.653121	-0.453121

In [33]:
```
df.head(15)
```

Out[33]:

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	
0	0.00632	18.0	2.31	0.000000	0.538	6.575	65.2	4.0900	1	296	15.3	396.
1	0.02731	0.0	7.07	0.000000	0.469	6.421	78.9	4.9671	2	242	17.8	396.
2	0.02729	0.0	7.07	0.000000	0.469	7.185	61.1	4.9671	2	242	17.8	392.
3	0.03237	0.0	2.18	0.000000	0.458	6.998	45.8	6.0622	3	222	18.7	394.
4	0.06905	0.0	2.18	0.000000	0.458	7.147	54.2	6.0622	3	222	18.7	396.
5	0.02985	0.0	2.18	0.000000	0.458	6.430	58.7	6.0622	3	222	18.7	394.
6	0.08829	12.5	7.87	0.069959	0.524	6.012	66.6	5.5605	5	311	15.2	395.
7	0.14455	12.5	7.87	0.000000	0.524	6.172	96.1	5.9505	5	311	15.2	396.
8	0.21124	12.5	7.87	0.000000	0.524	5.631	100.0	6.0821	5	311	15.2	386.
9	0.17004	12.5	7.87	0.069959	0.524	6.004	85.9	6.5921	5	311	15.2	386.
10	0.22489	12.5	7.87	0.000000	0.524	6.377	94.3	6.3467	5	311	15.2	392.
11	0.11747	12.5	7.87	0.000000	0.524	6.009	82.9	6.2267	5	311	15.2	396.
12	0.09378	12.5	7.87	0.000000	0.524	5.889	39.0	5.4509	5	311	15.2	390.
13	0.62976	0.0	8.14	0.000000	0.538	5.949	61.8	4.7075	4	307	21.0	396.
14	0.63796	0.0	8.14	0.069959	0.538	6.096	84.5	4.4619	4	307	21.0	380.

In [34]:
```
regression.predict([[0.62976,0.0,8.14,0.0,0.538,5.949,61.8,4.7075,4,307,21.0,396.60
```

```
C:\Users\Ganesh\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWa
rning: X does not have valid feature names, but LinearRegression was fitted with fea
ture names
  warnings.warn(
```

Out[34]:
```
array([19.58009845])
```

In [35]:
```
regression.intercept_
```

Out[35]: np.float64(35.040166029487196)

In [36]:
```
regression.coef_
```

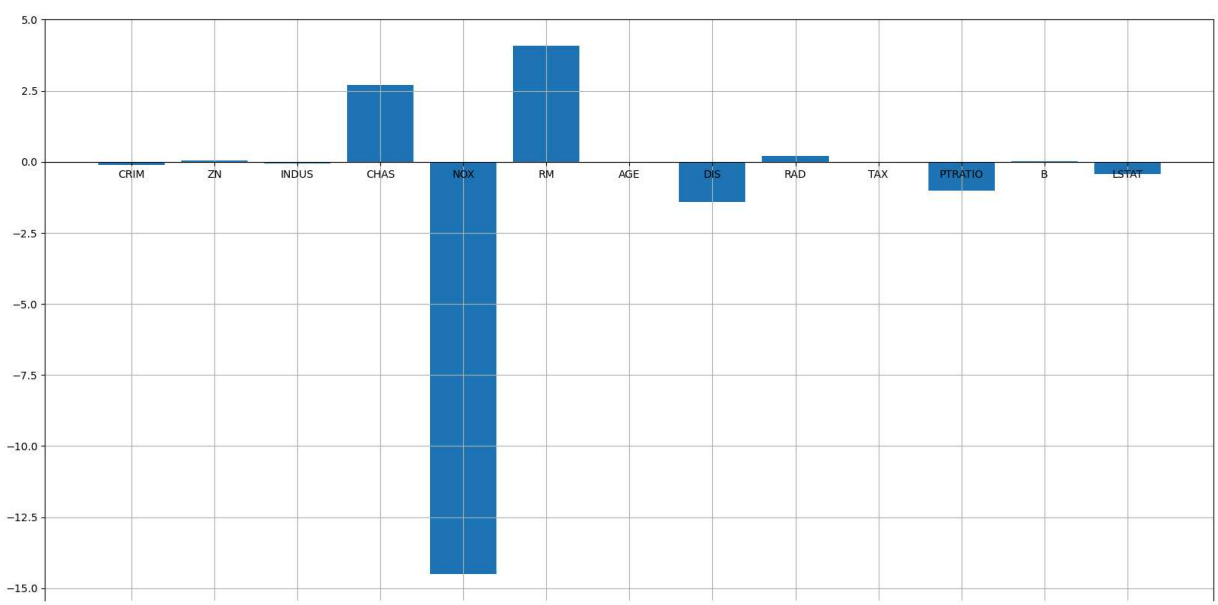
Out[36]:
```
array([-1.26194005e-01,  3.76363553e-02, -6.26295345e-02,  2.70382928e+00,
       -1.45015824e+01,  4.08006958e+00, -2.11509464e-02, -1.41798662e+00,
        1.96343241e-01, -8.70651696e-03, -1.01396225e+00,  8.29504244e-03,
       -4.19861039e-01])
```

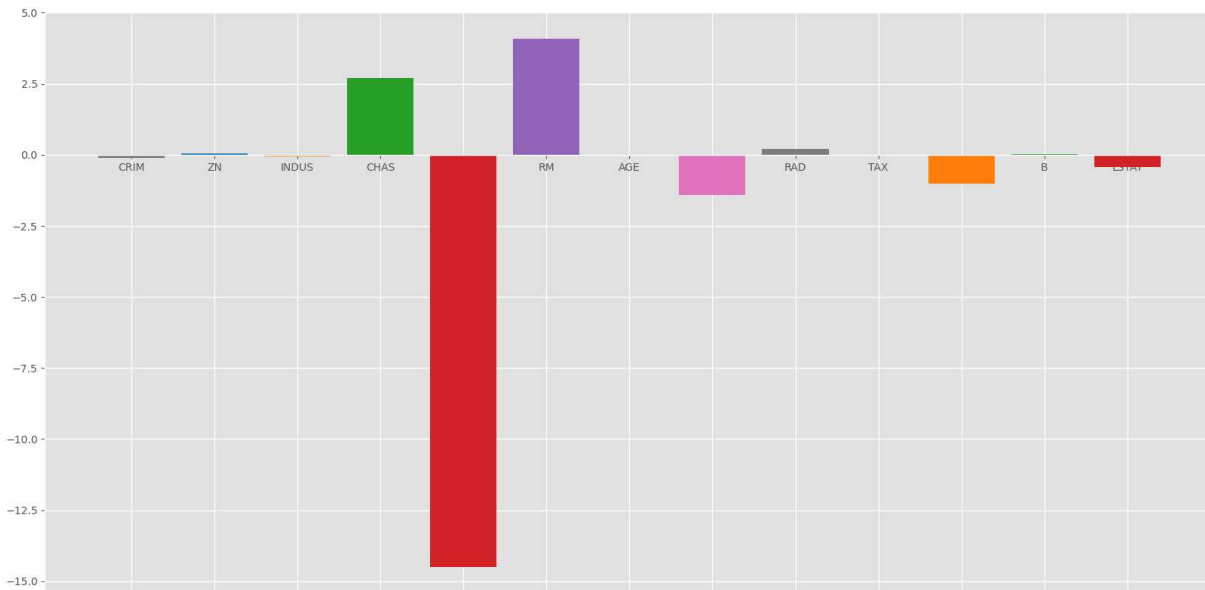
In [37]:
```
lr_coefficient = pd.DataFrame()
lr_coefficient["columns"] = x_train.columns
```

In [38]:
```
lr_coefficient['Coefficient Estimate'] = pd.Series(regression.coef_)
print(lr_coefficient)
```

```
   columns  Coefficient Estimate
0     CRIM             -0.126194
1       ZN              0.037636
2    INDUS             -0.062630
3     CHAS              2.703829
4      NOX            -14.501582
5       RM              4.080070
6      AGE             -0.021151
7      DIS             -1.417987
8      RAD              0.196343
9      TAX             -0.008707
10 PTRATIO             -1.013962
11        B              0.008295
12    LSTAT             -0.419861
```

In [39]:
```
fig, ax = plt.subplots(figsize =(20, 10))
ax.bar(lr_coefficient["columns"],
lr_coefficient['Coefficient Estimate'])
ax.spines['bottom'].set_position('zero')
plt.style.use('ggplot')
plt.grid()
plt.show()
```



In [ ]:

In [42]:
```
fig, ax = plt.subplots(figsize=(20, 10))
# Fixed the color list by properly closing the string for the last color
color = ['tab:gray', 'tab:blue', 'tab:orange', 'tab:green', 'tab:red', 'tab:purple'
ax.bar(lr_coefficient["columns"],
       lr_coefficient['Coefficient Estimate'], color=color)
ax.spines['bottom'].set_position('zero')
plt.style.use('ggplot')
plt.show()
```